Matrices solutions

D-E01 Dimensions and entrywise multiplication

Derivation and check

The product AB has dimensions 2×2. Its four entries are

$$
\begin{aligned}
(AB)_{11} &= 1 \cdot 2 + 2 \cdot (-1) + 0 \cdot 4 = 0, \\
(AB)_{12} &= 1 \cdot 1 + 2 \cdot 3 + 0 \cdot 0 = 7, \\
(AB)_{21} &= 0 \cdot 2 + 1 \cdot (-1) + 1 \cdot 4 = 3, \\
(AB)_{22} &= 0 \cdot 1 + 1 \cdot 3 + 1 \cdot 0 = 3.
\end{aligned}
$$

For BA, each row is a combination of the two rows of A:

Row	Calculation	Result
First	$2\,\mathrm{row}_1(A) + \mathrm{row}_2(A)$	$(2, 5, 1)$
Second	$-\mathrm{row}_1(A) + 3\,\mathrm{row}_2(A)$	$(-1, 1, 3)$
Third	$4\,\mathrm{row}_1(A)$	$(4, 8, 0)$

Equivalently, the nine entry calculations are

$$
\begin{aligned}
(BA)_{11} &= 2 \cdot 1 + 1 \cdot 0 = 2, & (BA)_{12} &= 2 \cdot 2 + 1 \cdot 1 = 5, \\
(BA)_{13} &= 2 \cdot 0 + 1 \cdot 1 = 1, & (BA)_{21} &= (-1) \cdot 1 + 3 \cdot 0 = -1, \\
(BA)_{22} &= (-1) \cdot 2 + 3 \cdot 1 = 1, & (BA)_{23} &= (-1) \cdot 0 + 3 \cdot 1 = 3, \\
(BA)_{31} &= 4 \cdot 1 + 0 \cdot 0 = 4, & (BA)_{32} &= 4 \cdot 2 + 0 \cdot 1 = 8, \\
(BA)_{33} &= 4 \cdot 0 + 0 \cdot 1 = 0.
\end{aligned}
$$

Thus

$$
AB = \begin{bmatrix} 0 & 7 \\ 3 & 3 \end{bmatrix}, \qquad BA = \begin{bmatrix} 2 & 5 & 1 \\ -1 & 1 & 3 \\ 4 & 8 & 0 \end{bmatrix}.
$$

The products cannot be equal because their dimensions differ. Multiplication sums over the shared dimension and retains the two outer dimensions.

For $x = (1, -1, 2)^T$, the two row dot products give

$$
Ax = \begin{bmatrix} 1 \cdot 1 + 2(-1) + 0 \cdot 2 \\ 0 \cdot 1 + 1(-1) + 1 \cdot 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.
$$

The column method gives the same result:

$$
Ax = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.
$$

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Matrices solutions (continued)

D-E02 Associativity and noncommutativity

Derivation and check

Let A be $m \times n$, B be $n \times p$, and C be $p \times q$, with entries in a field. For every $1 \le i \le m$ and $1 \le j \le q$,

$$\begin{aligned}((AB)C)_{ij} &= \sum_{k=1}^{p} \left(\sum_{\ell=1}^{n} a_{i\ell} b_{\ell k} \right) c_{kj} \\ &= \sum_{k=1}^{p} \sum_{\ell=1}^{n} a_{i\ell} b_{\ell k} c_{kj} \\ &= \sum_{\ell=1}^{n} a_{i\ell} \left(\sum_{k=1}^{p} b_{\ell k} c_{kj} \right) \\ &= (A(BC))_{ij}.\end{aligned}$$

The sums are finite, so reordering them is valid. This proves associativity for all compatible matrices.

Take the real symmetric matrices

$$S = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \qquad T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Their products are

$$ST = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}, \qquad TS = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}.$$

The product ST is not symmetric. In general, when A and B are real symmetric matrices of the same size, $(AB)^T = B^T A^T = BA$. Hence AB is symmetric if and only if $AB = BA$. Associativity changes parentheses while preserving factor order. Compatibility of one product does not guarantee a defined reverse product or equality of the two.

D-E03 Permutation direction

Derivation and check

Left multiplication by P selects the old rows in the order $3, 1, 2$. Right multiplication selects the old columns in the order $2, 3, 1$:

$$PM = \begin{bmatrix} 7 & 8 & 10 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \qquad MP = \begin{bmatrix} 2 & 3 & 1 \\ 5 & 6 & 4 \\ 8 & 10 & 7 \end{bmatrix}.$$

The transpose is

Matrices solutions (continued)

$$P^T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

The columns of P are the distinct orthonormal basis vectors e_2, e_3, e_1, so $P^T P = I_3$ and $P^{-1} = P^T$. Each row of a left factor selects an old row; each basis column of a right factor selects an old column. This explains the different directions. Here P is not its own inverse. The identities $P^3 = I_3$ and $P^2 = P^T$ give additional checks.

D-E04 Rank and the whole null space

Derivation and check

Each row operation gives the following complete state:

$$M = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivots are in columns 1 and 2. Select these columns from the original M, and record each original column's coordinates:

$$B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \qquad M = BC.$$

The third column is the sum of the first two. Set the free variable $x_3 = t$. The reduced equations give $x_1 = x_2 = -t$, so

$$\ker M = \{t(-1, -1, 1)^T : t \in \mathbb{R}\}.$$

A null-space basis is $\{(-1, -1, 1)^T\}$. The rank is 2 and the nullity is 1. The RREF pivot columns are e_1, e_2 in $\mathbb{R}^3$. Multiplying those columns by C reproduces R, rather than M. Original vectors and their coordinates after elimination must remain distinct.

Matrices solutions (continued)

D-E05 Product rank and invertibility

Derivation and check

Let $A \in \mathbb{F}^{m\times n}$ and $B \in \mathbb{F}^{n\times p}$ over a field $\mathbb{F}$. If $\operatorname{rank}(A) = r$, a rank factorization $A = UV$ has $U \in \mathbb{F}^{m\times r}$. Then $AB = U(VB)$, so every column of AB is in the span of the r columns of U, giving $\operatorname{rank}(AB) \le r$.

If $\operatorname{rank}(B) = s$, write $B = WX$ with $W \in \mathbb{F}^{n\times s}$. Now $AB = (AW)X$, giving $\operatorname{rank}(AB) \le s$. Together,

$$\operatorname{rank}(AB) \le \min\{\operatorname{rank}(A), \operatorname{rank}(B)\}.$$

The matrices $A = \operatorname{diag}(1, 0)$ and $B = \operatorname{diag}(0, 1)$ both have rank one, but $AB = 0$.

For an invertible $P \in \mathbb{F}^{m\times m}$, the product bound gives $\operatorname{rank}(PA) \le \operatorname{rank}(A)$. Applying it to $A = P^{-1}(PA)$ gives the reverse inequality, so the ranks are equal. Being nonzero does not suffice for P. Rank factorization and elimination use field arithmetic; conclusions over composite-modulus rings require separate justification.

D-E06 Complete inverse trace

Derivation and check

Start with the augmented matrix $[H \mid I_2]$. The first operation subtracts $3/2$ times row one from row two:

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - (3/2)R_1} \left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 0 & 1/2 & -3/2 & 1 \end{array}\right].$$

Normalize the second pivot:

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 0 & 1/2 & -3/2 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow 2R_2} \left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 0 & 1 & -3 & 2 \end{array}\right].$$

Eliminate above that pivot:

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 0 & 1 & -3 & 2 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1 - R_2} \left[\begin{array}{cc|cc} 2 & 0 & 4 & -2 \\ 0 & 1 & -3 & 2 \end{array}\right].$$

Normalize the first pivot:

$$\left[\begin{array}{cc|cc} 2 & 0 & 4 & -2 \\ 0 & 1 & -3 & 2 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1/2} \left[\begin{array}{cc|cc} 1 & 0 & 2 & -1 \\ 0 & 1 & -3 & 2 \end{array}\right].$$

Thus $K = H^{-1} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$. Both multiplication checks are explicit:

$$HK = \begin{bmatrix} 4-3 & -2+2 \\ 6-6 & -3+4 \end{bmatrix} = I_2, \qquad KH = \begin{bmatrix} 4-3 & 2-2 \\ -6+6 & -3+4 \end{bmatrix} = I_2.$$

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Matrices solutions (continued)

If B and C are both two-sided inverses of a square A, then $B = B(AC) = (BA)C = C$, proving uniqueness.

For the rectangular matrices $A = \begin{bmatrix} 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, one has $AB = I_1$ but $BA = \operatorname{diag}(1, 0) \neq I_2$. The square premise links injectivity with surjectivity and cannot be dropped.

D-E07 Cofactors and determinant rules

Derivation and check

The first-row minor matrices and their determinants are

Deleted column	Minor matrix	Determinant	Cofactor
1	$\begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$	$3 \cdot 1 - 2(-1) = 5$	5
2	$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$	$1 \cdot 1 - 2 \cdot 0 = 1$	-1
3	$\begin{bmatrix} 1 & 3 \\ 0 & -1 \end{bmatrix}$	$1(-1) - 3 \cdot 0 = -1$	-1

The cofactor signs are $+, -, +$, so $\det N = 2 \cdot 5 + 1(-1) + 0(-1) = 9$.

After $R_2 \leftarrow R_2 + R_1$, the matrix and determinant are

$$N' = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 4 & 2 \\ 0 & -1 & 1 \end{bmatrix}, \qquad \det N' = 2(4 + 2) - 3 = 9.$$

Swapping its first two rows gives

$$N'' = \begin{bmatrix} 3 & 4 & 2 \\ 2 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, \qquad \det N'' = 3 - 8 - 4 = -9.$$

For a general upper-triangular matrix, a nonzero term in the permutation expansion requires $\sigma(i) \geq i$ for every row i. Since a permutation satisfies $\sum_i \sigma(i) = \sum_i i$, every inequality must be an equality. Only $\sigma(i) = i$ survives, giving the diagonal product. For lower-triangular matrices the inequalities reverse and the same argument applies.

For a finite square matrix over a field, row swaps and nonzero row scalings multiply the determinant by nonzero factors, while row addition preserves it. Thus a zero determinant is equivalent to a missing pivot and hence to singularity. A nonzero exact determinant does not eliminate floating-point conditioning concerns.

Matrices solutions (continued)

D-E08 Outer-product boundaries

Derivation and check

For the specified vectors,

$$u^T v = 3 - 2 = 1, \qquad uv^T = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix}.$$

The columns are $3u$ and $-u$, so their span has dimension one. If $u = 0$, the outer product is the zero matrix and has rank zero.

Generally, for $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$, column j of uv^T is $v_j u$. Every column lies in $\operatorname{span}\{u\}$, so the rank is at most one. If both vectors are nonzero, some $v_j \neq 0$ gives a nonzero column that generates u, so the rank is exactly one.

Over $\mathbb{C}$, column j of uv^* is $\overline{v_j}u$, and the same rank condition holds. For equal-length complex vectors, the inner product is $u^*v = \sum_i \overline{u_i} v_i$, and $\|u\|_2^2 = u^*u \geq 0$. Ordinary transpose does not define the complex inner product.

D-E09 Gram and field counterexamples

Derivation and check

For X,

$$X^T X = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \begin{bmatrix} s & t \end{bmatrix} X^T X \begin{bmatrix} s \\ t \end{bmatrix} = s^2 + t^2 + (s+t)^2.$$

For every nonzero real pair (s, t), at least one of s^2, t^2 is positive, so the form is strictly positive.

For W,

$$W^T W = \begin{bmatrix} 5 & 10 \\ 10 & 20 \end{bmatrix}, \qquad \begin{bmatrix} s & t \end{bmatrix} W^T W \begin{bmatrix} s \\ t \end{bmatrix} = 5(s + 2t)^2.$$

This form is nonnegative but vanishes at the nonzero vector $(-2, 1)^T$, so it is positive semidefinite rather than positive definite.

For complex $A = [i]$, ordinary transpose gives $A^T A = [-1]$. Conjugate transpose gives $A^* = [-i]$ and $A^* A = [1]$. The claim has two missing premises. A nonzero matrix need not have full column rank, and complex Gram positivity requires conjugate transpose. Both the correct inner product and full column rank are required for strict positivity.

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Matrices solutions (continued)

D-E10 GF(2) maps and representation

Derivation and check

Over $\mathbb{F}_2$, multiplication gives $A(x_0, x_1)^T = (x_0 \operatorname{XOR} x_1, x_1)^T$. Retaining the integer representation $x = x_0 + 2x_1$ gives

Input integer	Input vector	Output vector	Output integer
0	$(0,0)^T$	$(0,0)^T$	0
1	$(1,0)^T$	$(1,0)^T$	1
2	$(0,1)^T$	$(1,1)^T$	3
3	$(1,1)^T$	$(0,1)^T$	2

Each integer occurs exactly once, so the map is a permutation. In $\mathbb{F}_2$, $1 + 1 = 0$, giving $A^2 = I_2$. Hence $A^{-1} = A$ and the rank is 2. Over the reals, the same displayed matrix instead satisfies

$$A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \neq I_2.$$

The arithmetic definitions change the result. Modulo 6, every integer $2y$ is even and cannot be congruent to 1. Thus 2 has no inverse despite being nonzero. The field guarantee that every nonzero scalar is invertible is essential for pivot normalization.

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Matrices solutions (continued)

Verification code appendix: matrix_checks.py

```
"""Exact Fraction matrix arithmetic checks; proof statements live in the text."""
from fractions import Fraction as F
from itertools import product
def mul(a,b):return [[sum(F(x)*F(y) for x,y in zip(row,col)) for col in zip(*b)] for row in
a]
def tr(a):return list(map(list,zip(*a)))
def reduce(a):
 a=[list(map(F,row)) for row in a];r=0;piv=[]
 for j in range(len(a[0])):
  t=next((i for i in range(r,len(a)) if a[i][j]),None)
  if t is None:continue
  a[r],a[t]=a[t],a[r];v=a[r][j];a[r]=[x/v for x in a[r]]
  for i in range(len(a)):
   if i!=r:
    v=a[i][j];a[i]=[x-v*y for x,y in zip(a[i],a[r])]
  piv.append(j);r+=1
 return a,piv
def det(a):
 if not a:return F(1)
 if len(a)==1:return F(a[0][0])
 return sum((-1)**j*F(a[0][j])*det([row[:j]+row[j+1:] for row in a[1:]]) for j in
range(len(a)))
if __name__=='__main__':
 a=[[1,2,0],[0,1,1]];b=[[2,1],[-1,3],[4,0]]
 assert mul(a,b)==[[0,7],[3,3]] and mul(b,a)==[[2,5,1],[-1,1,3],[4,8,0]]
 m=[[1,2,3],[2,4,6],[0,1,1]];r,piv=reduce(m);assert piv==[0,1] and r==[[1,0,1],[0,1,1],
[0,0,0]]
 assert mul([[1,2],[2,4],[0,1]],[[1,0,1],[0,1,1]])==m
 assert mul(m,[[-1],[-1],[1]])==[[0],[0],[0]]
 h=[[2,1],[3,2]];hi=[[2,-1],[-3,2]];assert mul(h,hi)==mul(hi,h)==[[1,0],[0,1]]
 assert det([[2,1,0],[1,3,2],[0,-1,1]])==9
 x=[[1,0],[0,1],[1,1]];w=[[1,2],[2,4],[0,0]]
 assert mul(tr(x),x)==[[2,1],[1,2]] and mul(tr(w),w)==[[5,10],[10,20]]
 assert mul(x,[[-2],[1]])==[[-2],[1],[-1]] and mul(w,[[-2],[1]])==[[0],[0],[0]]
 cases=0
 for flat in product(range(-1,2),repeat=4):
  q=[list(flat[:2]),list(flat[2:])];rr,pp=reduce(q)
  assert (len(pp)==2)==(det(q)!=0);assert len(pp)==len(reduce(tr(q))[1]);cases+=1
 assert [((z&1)^((z>>1)&1))+2*((z>>1)&1) for z in range(4)]==[0,1,3,2]
 print(f'PASS finite: {cases} exact2x2 rank/determinant/transpose identities; all chapter
numerical products, inverse, rank factors, Gram and GF2 mapping')
```

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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